



The role of asymmetric metal flow on weld formation and solidification characteristics during pulsed laser butt welding with assembly tolerance

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ABSTRACT

In comparison to other laser-assisted welding techniques, the pulsed laser welding process (PLBW) provides cost-effectiveness and ease of operation while effectively connecting butt-assembled thin plates with manufacturing tolerances (both air gap and edge misalignment). A combination of numerical simulations and experimental studies was conducted to analyze the forming mechanism and solidification behavior of the weld. In this paper, a transient three-dimensional heat-flow-metallurgical model has been established for a PLBW process of mild steel plates with a thickness of 1.8 mm. The predicted weld pool profiles were consistent with the experimental findings. The finding suggested the molten metal, driven by surface tension and recoil pressure, successively generated liquid bridges on the rear and front sides of the pool. The two sheets were joined as the molten metal gradually solidified. The energy absorptivity was highly correlated with the weld pool and keyhole dynamics, and the laser energy absorbed by the metal was mainly transferred in the form of thermal convection. The kinematic features of the melt were observed by the liquid phase mass transfer rate and surface fluctuations. According to the Fourier analysis results, the molten pool oscillated at the characteristic frequency of the laser pulse frequency. The spatial and temporal distributions of solidification parameters aided in analyzing the solidification features within the pool. The asymmetric flow had a greater influence on the temperature gradient. These findings addressed the roles of PLBW on fluid flow, heat behaviors, and solidification parameters of welds with assembly tolerances, as well as the reasons for improved weld formation.

1. Introduction

Laser technology has been widely employed in industrial manufacturing fields comprising laser welding [1], laser additive manufacturing [2], and laser drilling [3] due to its intensely concentrated energy and excellent efficiency. Among them, laser welding has emerged as an important means of connecting thin plates given that it can produce weld structures with low welding deformation as well as superior mechanical properties [4]. As the laser beam is highly focused, any assembly tolerance in a butt structure, such as air clearance and edge misalignment, can result in leakage and deviation of the laser energy, affecting the quality of the weld joint. Manufacturing tolerances, mostly caused by poor machining precision, assembly errors, and weld distortion, are unavoidable in actual production. As a result, the surface state of the buttress structure would change, as would the flow behaviors of the molten metal, which directly determine the forming state of the weld seams. Furthermore, the mechanical properties of the joint are

intimately tied to the solidification microstructure, which is considerably influenced by heat transfer and melt flow histories. Therefore, understanding the transient behaviors of heat transfer and melt flow within the molten pool in the presence of assembly tolerances, as well as their effects on solidification characteristics, is critical in analyzing the weld forming mechanism and controlling weld formation and performance.

The existence of assembly tolerances causes a partial loss of focused laser energy, which is reflected in the reduced bridgeability of the butt weld. Accordingly, the performance of the joint deviates from the ideal level [5]. To improve the sensitivity of the thin plate laser welding process to manufacturing tolerances, scholars have proposed several implementable schemes. Üstündağ et al. conducted a series of welding experiments using the hybrid laser and arc heat source in the case where manufacturing tolerances existed and summarized the maximum values of connectable gap bridgeability and edge misalignment [6]. Egerland et al. suggested a laser-metal active gas tandem hybrid welding

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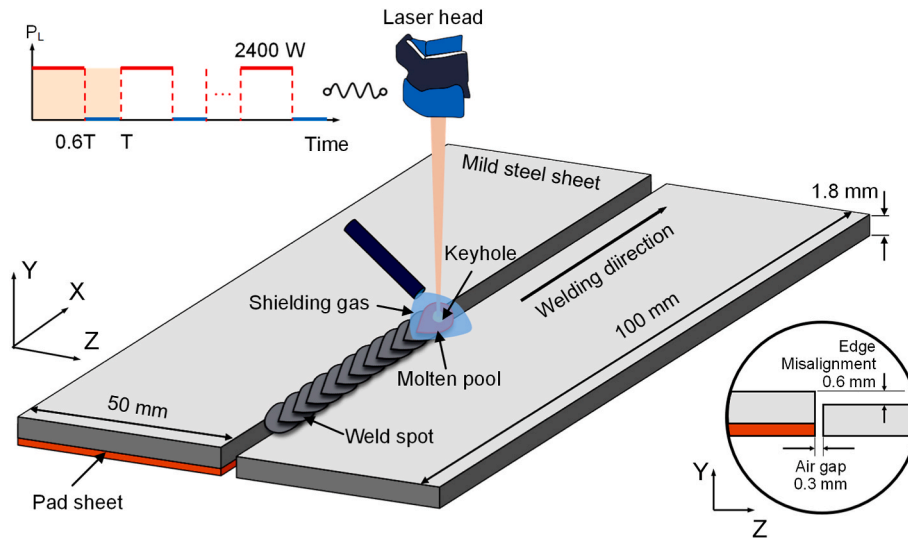


Fig. 1. Schematic diagram for mild steel PLBW process.

technology that controlled the welding speed by adjusting the wire diameter [7]. Sun and Kuo demonstrated that 2.0 mm thick butt-jointed steel plates with a gap of 1.0 mm could be successfully joined by wire filling [8]. Pu et al. quantified the effect of wire feed speed on the weld formation with a narrow gap, concluding that the wire feed speed determined droplet transition patterns by changing the relative location of the wire fusing part in the gap [9]. Yang et al. showed that making a transition through a liquid bridge rather than a globular transfer could effectively reduce incomplete fusion defects in narrow gap filler welding [10]. Schultz et al. employed an oscillating laser beam with filler wire to join 1 mm thick aluminum plates with a gap, and they announced the maximum allowable tolerances [11]. Köhler et al. adopted an oscillating laser beam to enlarge the connectable gap size from less than 0.1 mm–0.4 mm [12]. But in fact, these solutions were limited by increased needs for welding equipment or the installation of an extra post-welding machining sequence.

The author's team previously proposed employing a pulsed laser to deal with the working conditions where an air gap or edge misalignment existed [13,14]. The pulsed laser welding technology has the advantages of being both cost-effective and easy to implement. When compared to laser-MIG hybrid welding, laser fillet welding, and laser oscillating welding reported in the literature, it does not necessitate the use of extra equipment or machining operations. According to Tzeng et al. [15] and Torkamany et al. [16], the mean value of the maximum power density was the most valuable parameter attributed to the weld quality, and a pulse parameter called overlapping factor could be defined to help evaluate the melting ratio of pulsed laser process. Mohid et al. observed two melting zones with either higher laser power or larger pulse frequency [17]. Assuncao and Williams compared the thermal input in continuous mode laser and pulsed laser and found less energy was required for the latter to reach the same penetration depth [18]. PLBW process has a promising potential for reducing the formation of both intermetallic compounds [19] and thermal cracks [20]. Welded joints with small deformations have also been proven to be available via the PLBW technique [21]. By remelting more metal using a larger overlap factor, Gao et al. found the pores density in the weld seam was significantly decreased, because the bubbles had more chances to get away from the molten pool [22]. In addition, the pulsed laser set in the mode with a large spot as well as a low frequency offers impressive advantages in facilitating the forming of butt joints with assembly tolerances. First, the enlarged spot size on the focal plane directly expands the heated area of the workpiece. More importantly, the range of the laser beam with higher levels of energy is effectively widened, e.g., in the case of laser

beams with a top-hat shape, the top of the hat is stretched. Second, the low-frequency pulses (20–100 Hz) assist in ensuring the effective overlap of solder joints by regulating the overlapping factor.

Thanks to advances in computing capabilities, CFD tools are gradually being implemented to monitor variables that are difficult to measure directly, such as temperature and fluid velocity, and therefore to investigate the heat transfer and molten flow inside the welding pool. Cho and Na proposed a raytracing technique to realize the multiple reflections of the pulsed laser beams on the keyhole wall [23]. Wu et al. found the violent melt behaviors favored the occurrence of spatter in the laser welding pool for aluminum alloys [24]. Cho et al. simulated the interactions between the metal wire and the oscillating laser beam based on CFD theory and found the oscillation frequency had an important effect on both buttonhole formation and melt pool surface fluctuation [25,26]. Ke et al. systematically analyzed the thermal patterns and flow behaviors for three modes in the laser welding process with an oscillating laser beam [27]. Compared to atmospheric environments, Li et al. found the keyhole generated at subatmospheric pressure was wider and deeper with fewer bumps, while the melt pool was more stable [28]. Huang et al. reproduced the entire life cycle of bubbles from generation to capture by the solidification front in a hybrid laser-MIG welding process [29]. We have explained the formation mechanism of butted thin plates with a reserved gap based on the melt pool flow behaviors [13] and investigated the influence of the melt flow on the asymmetric geometries of both weld pool and keyhole in the presence of edge misalignment [14].

The service performance of welded joints is largely determined by the solidification organization. It is easier to analyze the transient solidification behavior by forecasting the microstructure based on the time-varying heat-flow state than by utilizing microstructure photography technologies. There have been some convincing studies on the relationship between heat transfer, flow pattern, and solidification morphology. Based on numerical simulations of three-dimensional heat transfer and fluid flow, Wei et al. investigated the solidification growth patterns of the nickel-based alloy under different laser scanning methods [30]. Ou et al. developed a computational framework that combines heat-fluid calculation and solidification parameters prediction for arc additive manufacturing processes. The results derived from different process parameters were compared [31]. Geng et al. studied the role of heat transmission and metal flow behaviors on the microstructure of aluminum alloy after welding solidification. Solidification characteristics such as temperature gradient and solidification rate, as well as the combining forms of the two, were evaluated [32]. Li et al. coupled a

Table 1
Chemical components of the base metal DC01EK (wt, %).

C	Si	Mn	P	S	Al	Fe
0.051	0.021	0.20	0.019	0.005	0.03	Bal.

metallurgical model with a thermal-fluid model for 304 stainless steel laser welded joints and obtained the contour results of solidification parameters [33]. Roehling et al. compared 316 L stainless steel cubes formed under a Gaussian-like beam and an elliptical-like beam. The latter was more advantageous in terms of thermal profile and microstructure [34]. Zhao et al. analyzed the impact of frequency in laser wobbling welding on the microstructure morphology and explored the reasons for grain refinement [35].

From the literature summarized above, several studies on the melt flow behaviors when air gaps or mismatched edges existed, as well as research on the solidification microstructure prediction using the thermal-fluid model, have been published. However, the forming mechanism for butt joints with both groove clearance and mismatched edges using PLBW is unclear, and the solidification parameters in the presence of assembly tolerances have seldom been reported. Thus, this study is conducted on the heat transfer, metal flow as well as solidification characteristics of the weld joints with assembly tolerances under the pulsed laser. The goal of this paper is to achieve an effective connection of a pair of mild steel plates butt-assembled with an air gap at 0.3 mm and an edge misalignment at 0.6 mm using the pulsed wave laser, which is assessed by examining the welding formation and mechanical properties. Additionally, the forming mechanism of butt welds with assembly tolerances is investigated based on the analysis of heat transfer and fluid flow. We also focus on the effects of heat-flow histories on solidification parameters. We hope this paper could be beneficial for comprehending the kinetic behaviors of the weld pool under the pulsed wave laser and the formation mechanism of the weld seam in the presence of assembly tolerances.

2. Methodology

2.1. Experimental setup

The pulsed laser welding experiments were carried out based on a laser remote welding equipment, as illustrated in Fig. 1. The platform was outfitted with a Trumpf TruDisk 8001 disc laser device having a maximum output of 8 kW, a KUKA robot arm and a PFO welding head. The laser waveform was modulated as a square wave and the peak power was 2.4 kW. A pair of mild steel plates (DC01EK) with dimensions of $100 \times 50 \times 1.8$ mm were clamped with an air gap of 0.3 mm and an edge misalignment of 0.6 mm. The chemical composition of the base metal is given in Table 1. The entire welding operation was conducted in a pure argon environment. The relevant experimental parameters are listed in Table 2. After the welding experiment, the molten pool geometry profile, metallographic microstructure and mechanical properties of the welded joints were observed and analyzed.

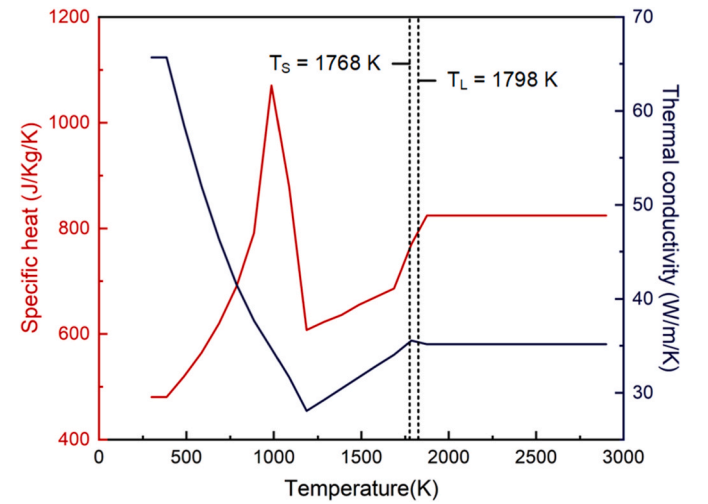


Fig. 3. Temperature-dependent parameters for heat-flow simulations.

Table 2
Experimental parameters for PLBW.

Gap size	Misalignment scale	Focused spot diameter	Laser scanning speed	Defocus distance	Pulse frequency	Duty cycle
0.3 mm	0.6 mm	0.7 mm	20 mm/s	0	20 Hz	0.6

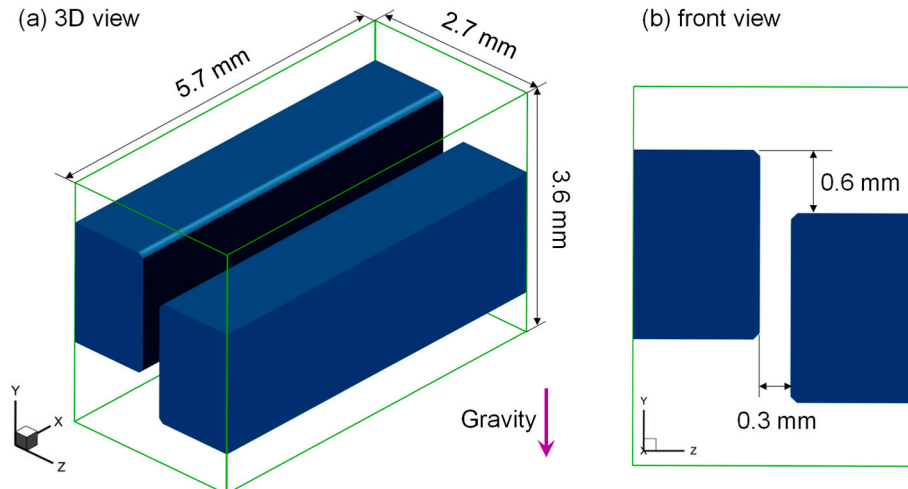


Fig. 2. Computing domain for CFD model.

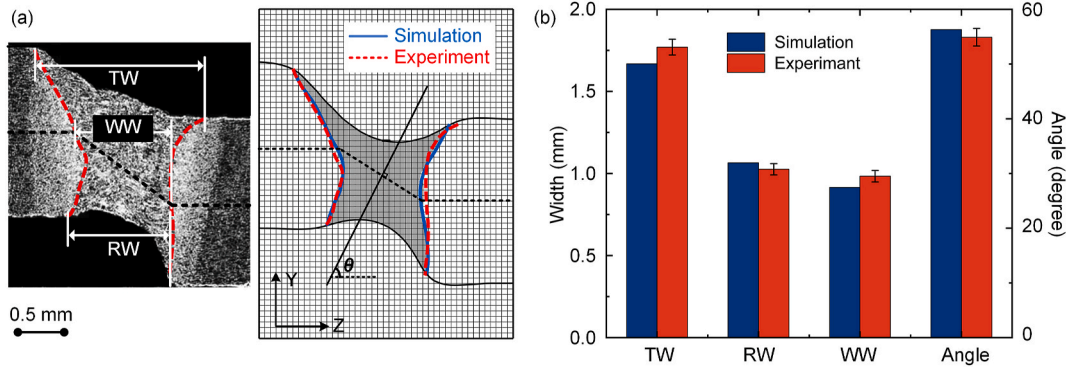


Fig. 4. Comparison of calculated weld pool profile and experimental one: (a) geometric contours; (b) measured dimensions.

2.2. CFD model

CFD calculation of pulsed laser welding with pre-assembly air gap and weld misalignment has been carried out to obtain melt flow behaviors and heat transfer characteristics within the weld pool. Fig. 2 shows the computational domain for this work. Considering the groove clearance size of 0.3 mm, the cell size was set to 0.06 mm, and the total number of meshes reached 256,500. The thermophysical properties of the mild steel and pure argon employed in the thermal-flow simulation can be found in our previous work [14]. Here, thermal conductivity and specific heat for steel are treated to be temperature-dependent, as drawn in Fig. 3.

A multi-phase incompressible laminar flow model for Newtonian fluid was established in the simulation. The LS-VOF coupling algorithm was adopted to capture the free surface of the melt pool, while the metal evaporation effect was equated to the movement of the gas-liquid interface. Conservation equations of mass, momentum, and energy were simultaneously solved to obtain spatial and temporal distributions of velocity, pressure, and temperature. The mathematical model for laser welding is described below. The focused laser beam was divided into 70×70 sub-beams whose energy density obeyed Gaussian distribution. The raytracing technique was utilized to realize multiple reflections of laser beams, in which the energy loss caused by the gas phase was considered as the inverse bremsstrahlung absorption and the energy absorption by metal as the Fresnel absorption. To characterize the scattering of the focal spot as well as the reheating impact of the plasma on the plate surface, a Gaussian-distributed heat source with a height of about three grids was applied to the top surface of the melt pool. Recoil pressure, buoyancy, surface tension and resistance force in the mushy zone were considered as the source terms of the momentum equation. The above equations were discretized based on the finite volume method and solved by Ansys. Fluent 2019R2 software. It took about 150 h to simulate the pulsed laser welding process with a duration of 0.15 s when using an Intel(R) Core(TM) i7-12700KF processor and 32 GB of RAM. The relevant details of the control equations for laser welding [13,14] are available in Appendix A.

3. Results and discussion

3.1. Model validation

To validate the reliability of the CFD model in this paper, the molten pool geometries on the weld cross-section for both the experimental and simulated results were compared. The interest slices were fixed at 3.1 mm from the weld starting point, which corresponded to the last moment of the second laser duration. The calculated profile of the weld pool was in high agreement with the metallographic finding, as shown in Fig. 4(a). Further comparisons were performed for the top weld width (TW), root weld width (RW), waist width (WW) and the inclination

Table 3

Simulated and experimental values of weld seam characteristic dimensions.

Data	TW (mm)	RW (mm)	WW (mm)	θ (degree)
Numerical	1.67	1.07	0.92	56.3
Experimental	1.77	1.03	0.98	54.6
Deviation (%)	-5.6	3.9	-6.1	3.1

angle of the normal line to the mid-thickness plane (θ), displayed in Fig. 4(b) and Table 3. The estimated dimensional values matched with the experimental observations well, with deviations varying from -6.1% to 3.9%, proving this model was reasonable.

3.2. Liquid bridge formation under reserved air gap and weld misalignment

The iso-surface with vof = 0.5 in the three-dimensional view is plotted in Fig. 5, showing the evolution of the molten pool when manufacturing tolerances exist. The temperature distribution is displayed in colors on the workpiece surface, with the solidus temperature outlined in black. As welding began, a shallow semicircular crater was left on each plate. The craters were minor in size due to the presence of an air gap that allowed most of the laser energy to escape, which was consistent with the results reported in Refs. [13,36]. As the laser energy continued to radiate, the weld pool top surface steadily expanded. The semi-pool on the right workpiece was somewhat smaller since it was clamped lower with more energy loss before the laser beam reached its surface. The liquid metal progressively bulged in response to surface tension and recoil pressure, and first formed a liquid bridge at the rear part of the pool. The two semi-pools at the front part were then also linked. Thereafter, a quasi-steady keyhole developed gradually, which had no difference from that in full-penetration laser welding under general conditions [37]. When the laser was switched off, the keyhole wall quickly surged back into place and two pieces of plates became united as the molten metal solidified, in keeping with the observations of Chen et al. [38]. The melt pool was a unit in the subsequent pulse cycle, without the step of joining two separated portions into a whole through a liquid bridge. In the next pulse cycle, laser radiation was emitted on the hardened solder junction due to the existence of the overlapping factor [16]. One appealing feature was the weld pool geometry gradually shifted towards the right side from the state where the left side dominated, as shown in Fig. 5 from 12 ms to 30 ms. For a more quantitative study, the weld widths on the top and bottom surfaces of two semi-pools were extracted, as shown in Fig. 6. Within the first pulse width stage, the melt width of the top surface on the higher-clamped side first grew to a peak value, then fell and stabilized to a certain level. The peak occurred within a few milliseconds after reaching the bilaterally bridged state. Besides, a comparable slight alteration was noticed on the lower side. These were somewhat different from the

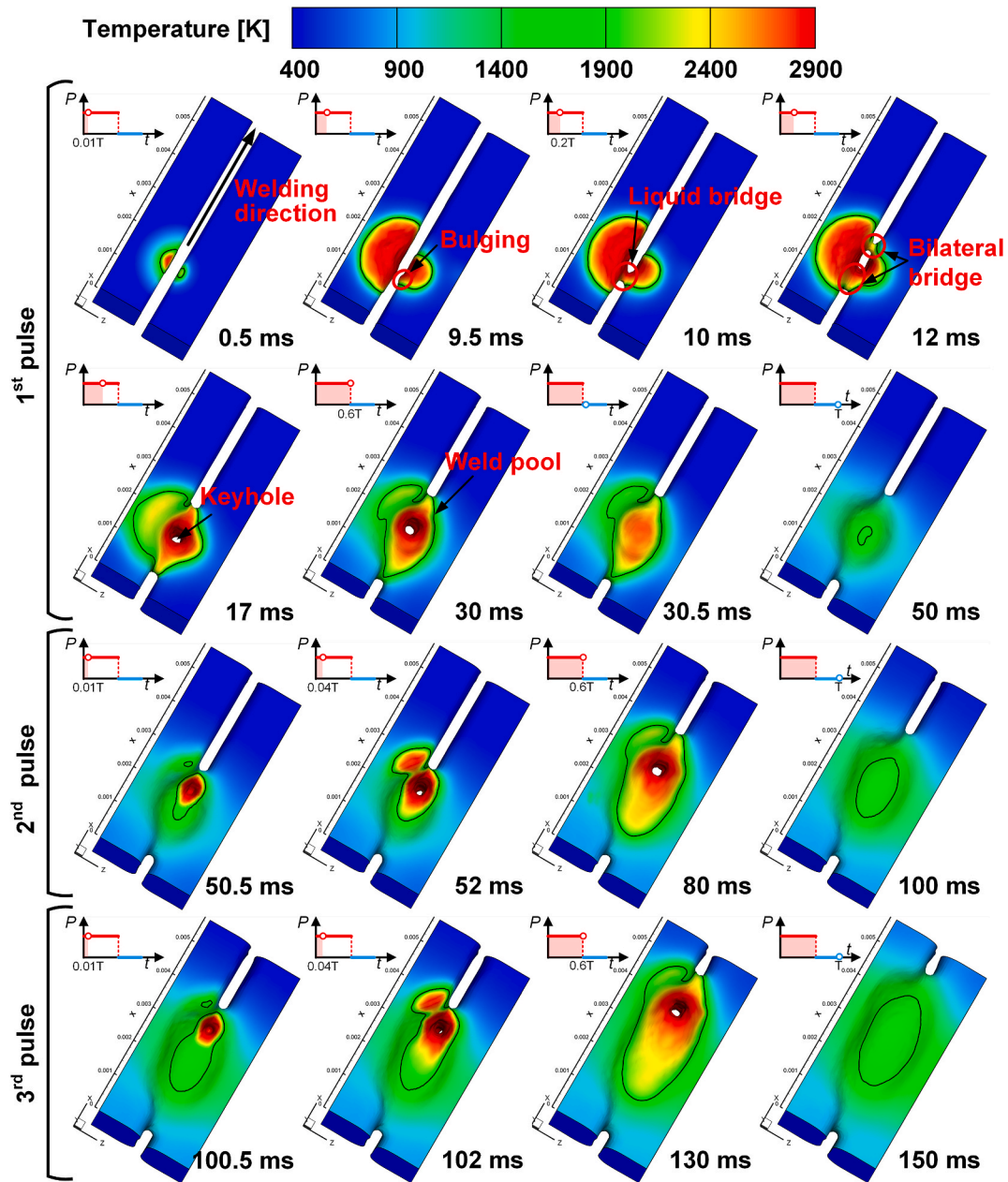


Fig. 5. Evolution of the melt pool bridging in the case of assembly tolerance.

results observed with only the weld misalignment [14]. The analysis should be performed in conjunction with the molten pool states in the cross-sections, as in Fig. 7, where the fluid velocity vectors are indicated by short straight arrows and the flow directions are represented by thick black curves with arrows. Before a complete liquid bridge was formed, the metal sheet on the left side got a larger melt pool surface since it received more laser energy than the right piece. Simultaneously, the left-bottom part of the workpiece was not yet melted because it relied only on the heat conduction within one piece of sheet. After the formation of the liquid bridge, the two pieces were connected as one. The stronger heat transfer assisted in melting the metal at the left-bottom side. The formation of the united pool considerably enlarged the heat transfer range, resulting in a reduction in the top surface weld widths on both assembled sides. In particular, a gravity-driven downward flow developed in the left pool, partially counteracting the Marangoni flow near the top surface. The asymmetry of the weld width on the root surface might be ascribed to greater heat accumulating on the left-side

metal sheet.

To clarify, the contours of the liquid phase were plotted in Fig. 8 in the weld midline plane which was full of mixed gas before welding. The red curve denotes the solidus temperature and the arrows indicate the flow direction. Not until 10 ms after welding began did a small piece of molten metal appear on the interested slice. After another 2 ms, a bilateral liquid bridge was obtained. The molten pool and keyhole then gradually stabilized, consistent with Fig. 5. The liquid metal was driven by the surface tension gradient to generate Marangoni flows on the top and root surfaces of the liquid pool. Then a distinct vortex formed as a result of drag force from a mushy zone. The two opposing vortices collided at mid-depth and canceled each other out, creating a circumferential flow towards the keyhole wall, confirming the arguments of Feng et al. [37]. The strong Marangoni effect facilitated heat transmission outward to create a greater melting volume, resulting in a continually growing liquid bridge. As the pulsed laser switched off and the weld pool size shrank, the metal was retained in the solid state on the

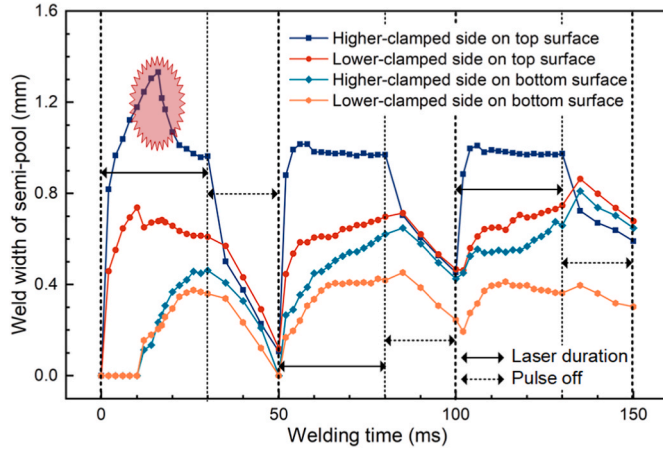


Fig. 6. Melt widths on top and bottom surfaces of two semi-pools.

analyzed plane, indicating the weld was formed. In later pulse cycles, the laser beam acted on the end part of the molten pool left from the previous cycle, re-forming the molten pool and keyhole. Along with energy transmission and laser beam movement, more metal was fused to enlarge the liquid bridge, which gradually solidified after the laser beam was turned off.

The area of the liquid bridge in the weld midline plane was measured and plotted versus welding time in Fig. 9. During the laser radiation period, the area tended to grow. In the second and third cycles, it only decreased slightly at the beginning of the pulses, caused by the formation of keyholes on the previously solidified metal. After the laser turned off, the liquid bridge area gradually stabilized because no energy input to melt additional metal.

3.3. Molten pool dynamic behaviors and heat transfer

The molten pool dynamic behaviors are intimately connected to the welding thermal process. In PLBW, laser energy absorptivity is an essential parameter. The energy from each sub-beam to metal sheets via Fresnel absorption has been tracked and added up. The ratio of total absorbed energy to incident value is therefore described as absorptivity [39], and its evolution with welding time is illustrated in Fig. 10. The number of multiple reflections used in this paper is fixed at three. The analysis is as follows when combined with the gas-liquid phase interface distribution in Fig. 5. At the starting point of the first pulse, only a tiny percentage of the sub-beams interacted with the sheets while the majority of the laser energy leaked from the air gap as the two pieces remained out of contact with each other. If the energy density within the focused spot was expressed in terms of Gaussian distribution, the captured sub-beams carried minimal energy since they were far from the spot center. As a result, the energy absorptivity was exceedingly low throughout this period. As the liquid bridge appeared and the molten pool reached a dynamic steady state, the energy absorption rate progressively raised and fluctuated around a certain value. Despite this, the overall level of energy absorptivity at this stage was lower than in the unpenetrated condition due to the presence of the through keyhole [40]. Additionally, the edge misalignment made the keyhole to be asymmetric about the laser axis at an oblique angle. This posture would affect the absorption of laser energy to some extent, as evidenced by the incident angle and reflection behaviors on the keyhole wall.

The Peclet number, Pe , aids in evaluating the heat transfer pattern inside the molten pool, as given by Eq. (1) [41].

$$Pe = \text{heat}_{\text{convection}} / \text{heat}_{\text{conduction}} = u\rho C_p L_R / \lambda \quad (1)$$

where, u is the characteristic flow velocity in weld pool, L_R means the characteristic length, ρ , C_p , λ have been defined earlier. Pe number

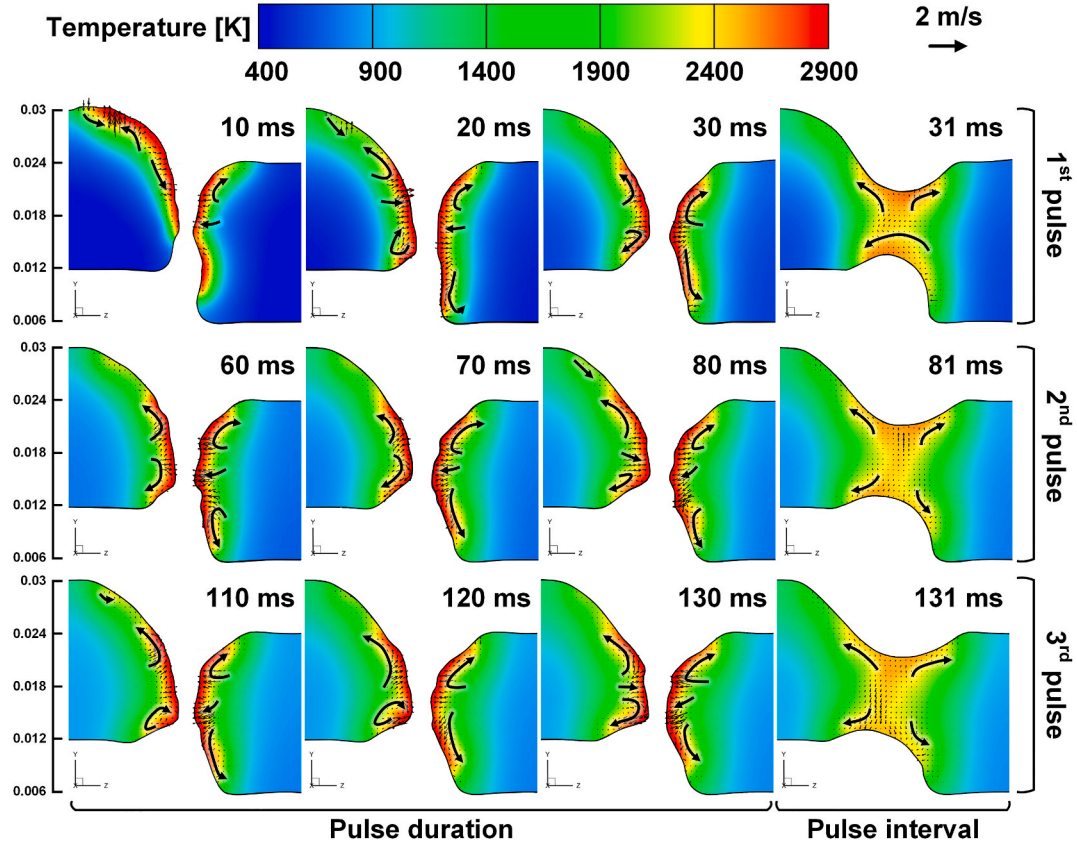


Fig. 7. Molten pool flow behaviors on weld cross-sections at different moments.

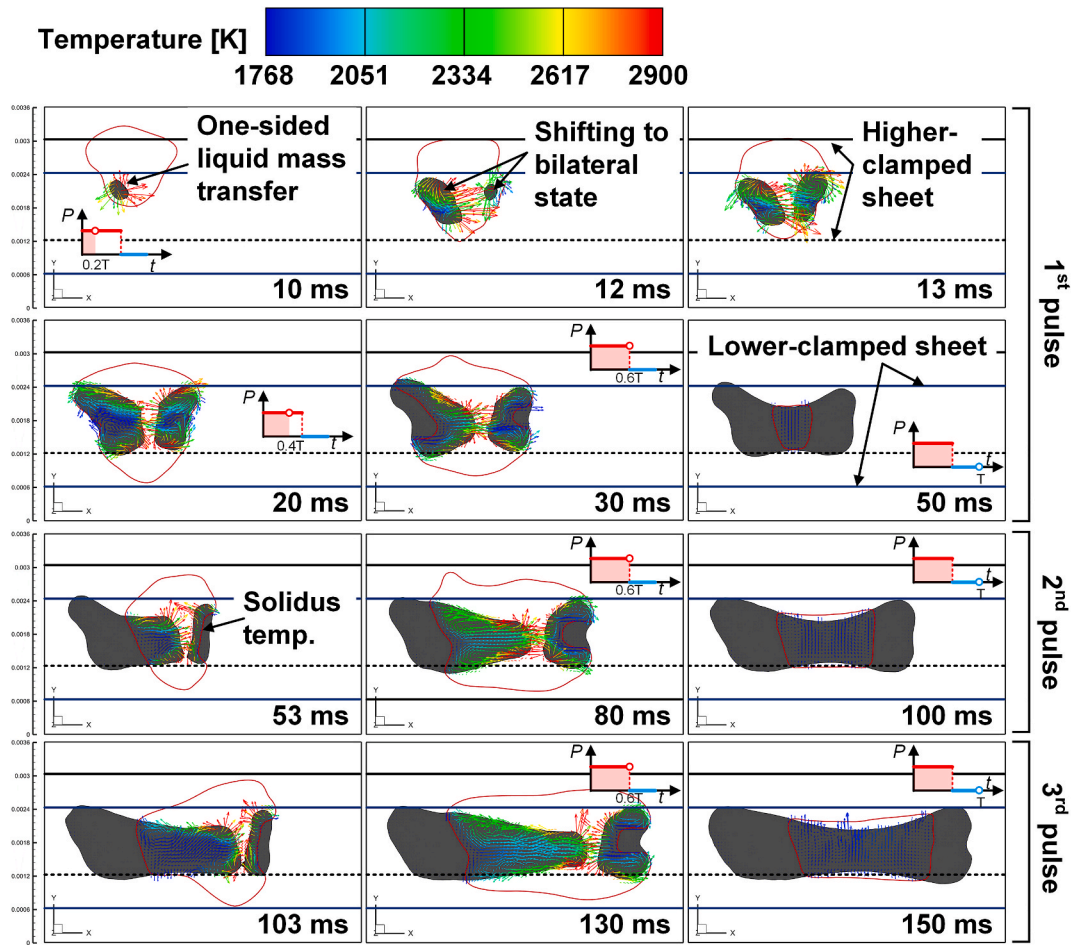


Fig. 8. Liquid bridge on longitudinal section of the weld.

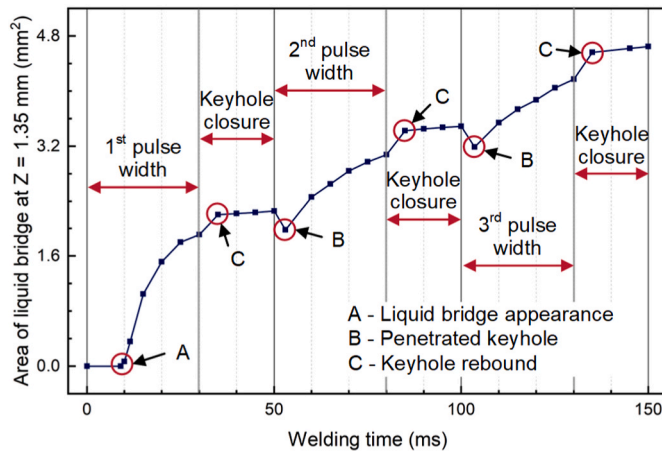


Fig. 9. Area of liquid bridge on weld longitudinal section versus welding time.

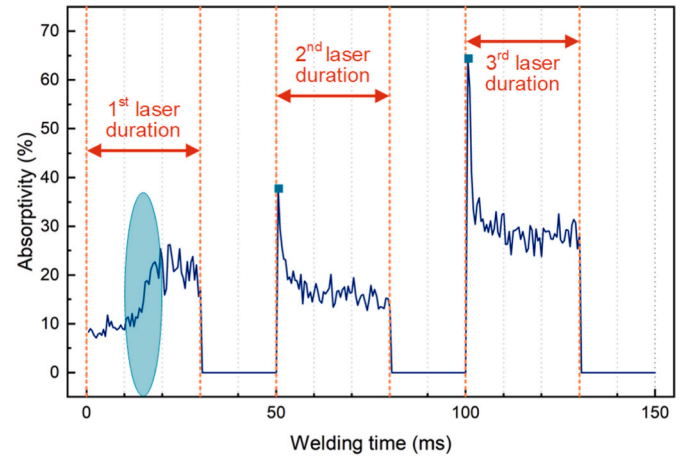


Fig. 10. Variation of laser energy absorptivity with welding time.

performs as a function of temperature since parameters C_p, λ are temperature-dependent. At the same time, different positions inside the molten pool will give different Pe numbers due to varying fluid velocities. For example, the Pe number near the weld pool surface is larger. We considered the characteristic fluid flow rate as 0.1 m/s and the characteristic length as 0.9 mm, which was half the weld width on the top surface. Parameters ρ, C_p, λ were given by Fig. 3, where C_p was taken as its minimum value of 400 J/kg/K and λ the maximum value of 70 W/m/K. The Pe number of 3.5 was obtained, indicating the dominance of

heat convective in the heat transfer pattern of the butt PLBW melt pool with air gap and edge misalignment. This was also confirmed by temperature contours as well as flow patterns, where the recirculating flow was apparent and had an important effect on the weld pool profile.

As illustrated in Fig. 11(a), the approximate mid-plane along the thickness direction of the asymmetric weld pool was chosen as the reference plane and colored in red. The real-time liquid phase mass flow rate through the reference plane was monitored and plotted in Fig. 11 (b), which facilitated to reveal the weld pool dynamic behaviors in the

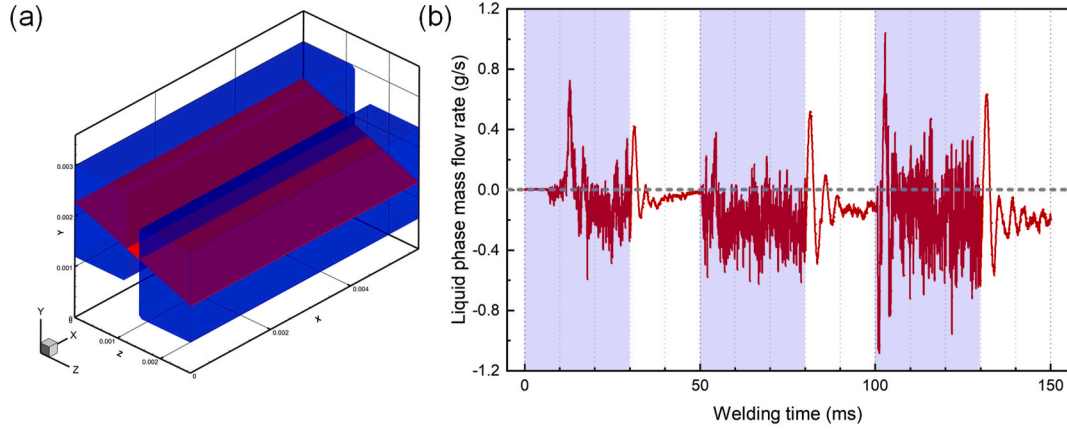


Fig. 11. Liquid phase mass transfer rate for a given plane in the metal pool.

vertical direction [42]. The mass flow rate varied from -0.87 g/s to 0.47 g/s during the first three pulse cycles. The graph showed that the melt pool oscillation was much more vigorous during laser radiation stage than the pulse interval as the result of more intense thermal interactions. The negative values of mass flow rate occupied more moments over the laser duration, which was associated with the gravity-driven flow to the lower-assembled side of the united melt pool. Such a considerable amount of negative values first appeared between 10 ms and 20 ms, corresponding to the moment in Fig. 5 when the bilateral liquid bridge was formed. As seen in Figs. 7 and 8, a strong upward flow occurred in the molten pool soon after the laser was switched off, which was a compensating current generated by the rapid rebound of the keyhole wall. The curve amplitude then swiftly dropped and the frequency of oscillation diminished, indicating a slight fluctuation inside the metal pool at this stage.

Temporal fluctuation motions of the melt pool surface were measured directly to characterize the melt pool dynamics. The sampling method [26] was shown in Fig. 12(a), with the measurement positions being at the weld centerline, 0.5 mm behind the beam center on the X-axis, and a sampling frequency of 2 kHz. The sampling data were plotted in Fig. 12(b) and (c). The top surface of the molten pool waved at 1.981 mm on average, which was below the top surface of the lower-clamped sheet. The bottom surface fluctuated at 1.312 mm on average, which was above the root surface of the higher-assembled workpiece. These undulations indicated the inward concave geometry of the molten pool without wire filler. The time measurements were converted to the frequency domain using the fast Fourier transform (FFT) technique, as in Fig. 12(d) and (e). In the frequency amplitude data, a characteristic frequency component with a value of 20 Hz shone out, which happened to be the laser repeat frequency. The weld pool fluctuating behaviors in PLBW were shown to be tightly related to the pulse frequency.

3.4. Solidification parameters

Numerous studies have found the morphology and scale of the weld microstructures are closely associated with the solidification parameters consisting of temperature gradient, solidification growth rate, and their combinations. The temperature gradient G could be calculated by Eq. (2) [43].

$$G = \sqrt{G_x^2 + G_y^2 + G_z^2}, \quad (2)$$

where, G_x , G_y , G_z denote the gradients of temperature in x , y , z directions, respectively.

The solidification growth rate R , also known as the solid-liquid interface movement velocity, is approximated as a projection of the

welding velocity in the normal direction of the solid-liquid front [32].

$$R = V \cdot i \cdot N_{sl}, \quad (3)$$

where, V is the laser scanning velocity, i means the unit vector in welding direction, N_{sl} is the normal vector of solid-liquid interface.

The parameter GR , known as the cooling rate, determines the coarseness of the microstructure. Generally, a large GR value produces a fine solidified microstructure. The morphology parameter G/R , which characterizes the supercooling degree, determines the morphology of the solidified microstructure. Different ratios make the weld morphology vary between planar, cellular, columnar and equiaxed structures. The solidification front destabilization criterion is stated in Eq. (4) [44].

$$G/R < \Delta T_E / D_L, \quad (4)$$

where, ΔT_E is 30 K, denoting the temperature difference from solidus temperature T_S to liquidus temperature T_L of mild steel DC01EK, D_L means the diffusivity of carbon in the liquid iron, which is about $4.79 \times 10^{-3} \text{ mm}^2/\text{s}$ [45].

According to Fig. 13, G and R were calculated at the isotherm $T = (T_L + T_S)/2 = 1783 \text{ K}$ on the horizontal plane. The variations of the solidification characteristic parameters on the horizontal planes at different thickness positions were extracted as Fig. 14. Since the normal of the isothermal profile did not coincide with the welding direction, a smaller R value existed at a larger angle between the two, which was in agreement with the published findings [32,35]. At the fusion line boundary, R approached zero because the normal vector of liquidus isotherm line was perpendicular to the weld direction. Near the weld center, R reached its maximum value as the normal line paralleled the weld direction. When comparing the R curves at three different Y values, the tilt towards the lower-assembled side of the whole metal pool could be confirmed again by the non-coincidence of the curve symmetry axes. Moreover, the R values on the upper and lower surfaces were slightly higher than the values at the mid-thickness plane, with the maximum value appearing near the molten pool top surface at about 20 mm/s, representing the welding speed. Typically, the G value fell drastically from the fusion boundary to the centerline, correlating to the distance variations between the solidus and liquidus isotherm lines [32]. Under the operating condition in this paper, G failed to behave in this pattern. The two probable explanations are displayed below. For starters, the welding speed of 0.02 m/s used in this research is quite moderate, resulting in a thin melt pool trailing scale. As a result, the distance between the solidus and liquidus isotherms is quite narrow and highly sensitive to changes in curve positions. For another, the asymmetry pool caused by the buttress misalignment significantly affected the G distribution, exhibiting a disproportion on the two sides at the position of $Z =$

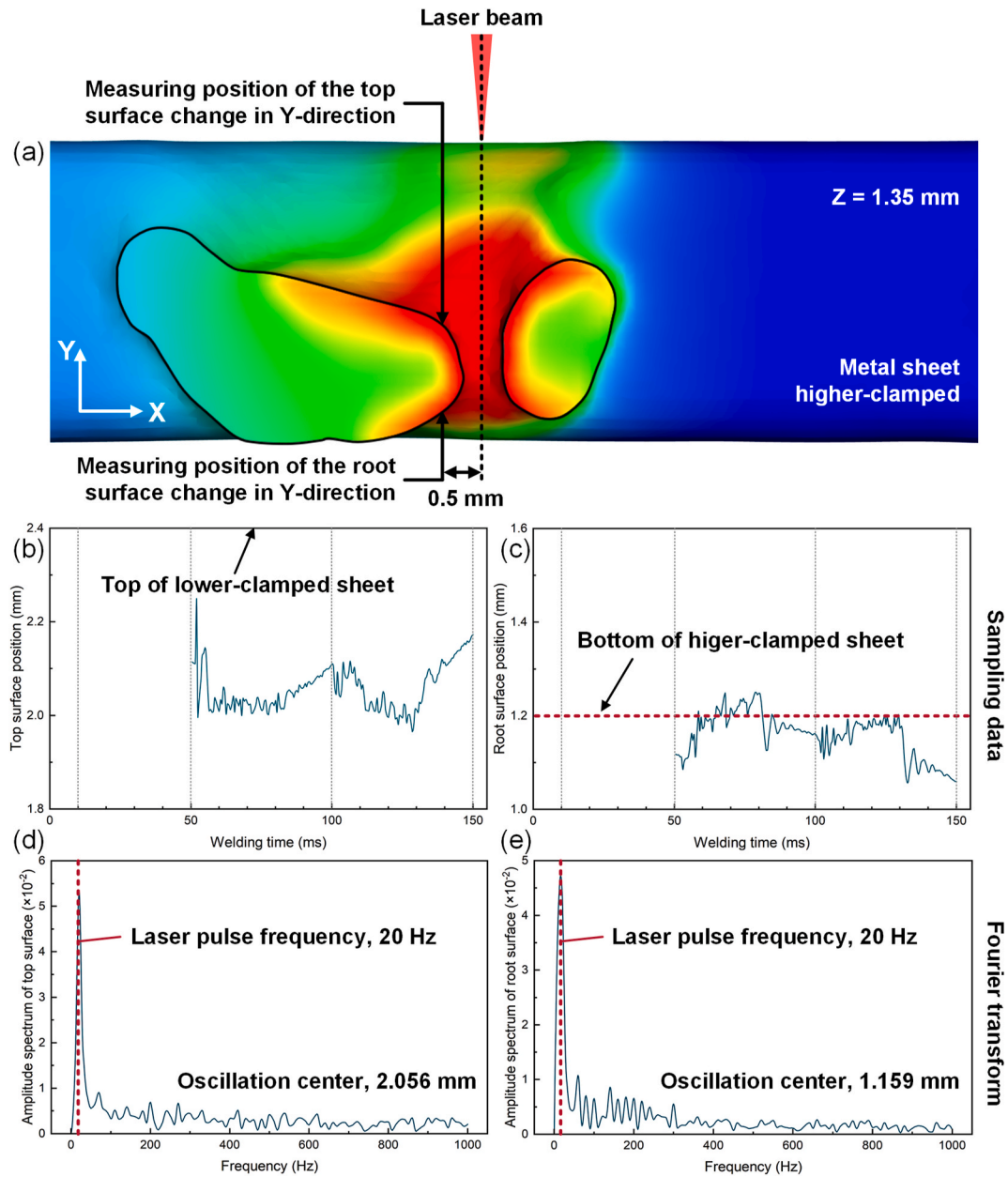


Fig. 12. Oscillations of molten pool surfaces: (a) measurement positions; temporal signals of (b) top surface and (c) root surface; frequency amplitude data of (d) top surface and (e) root surface.

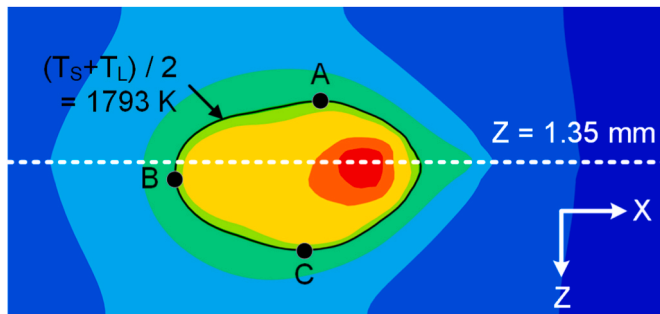


Fig. 13. Schematic diagram of sampling locations for G and R.

1.35 mm. When looking at the G curves for $Y = 1.4 \text{ mm}$ and $Y = 2.2 \text{ mm}$, it was easy to notice the trends on both sides of the assembly centerline were nearly mirrored, indicating the opposing flows that dominate in the upper and lower semi-pools. G varied from about 900 K/mm to 2800 K/mm in the sampling range. The G/R curves in three height planes, illustrated in Fig. 14 (c), dropped greatly from the fusion boundary to the centerline. The maximum value of G/R in the weld pool (about 1600 ks/mm²) was substantially lower than the value of $\Delta T_E/D_L$ for mild steel DC01EK (6260 ks/mm²). Therefore, the plane front destabilization condition in Eq. (4) was matched for the process parameters considered. Only cellular and dendritic structures would exist in the solidified weld. The supercooling degree GR at the solid-liquid boundary for three heights all grew progressively from the fusion boundary to the centerline, as in Fig. 14 (d). When Y moved from 1.4 mm to 1.8 mm–2.2 mm, GR at the weld pool centerline was 33,944 K/s, 3,3003 K/s and 41,932 K/s, respectively.

G and R were further investigated at the intersections of cross-section

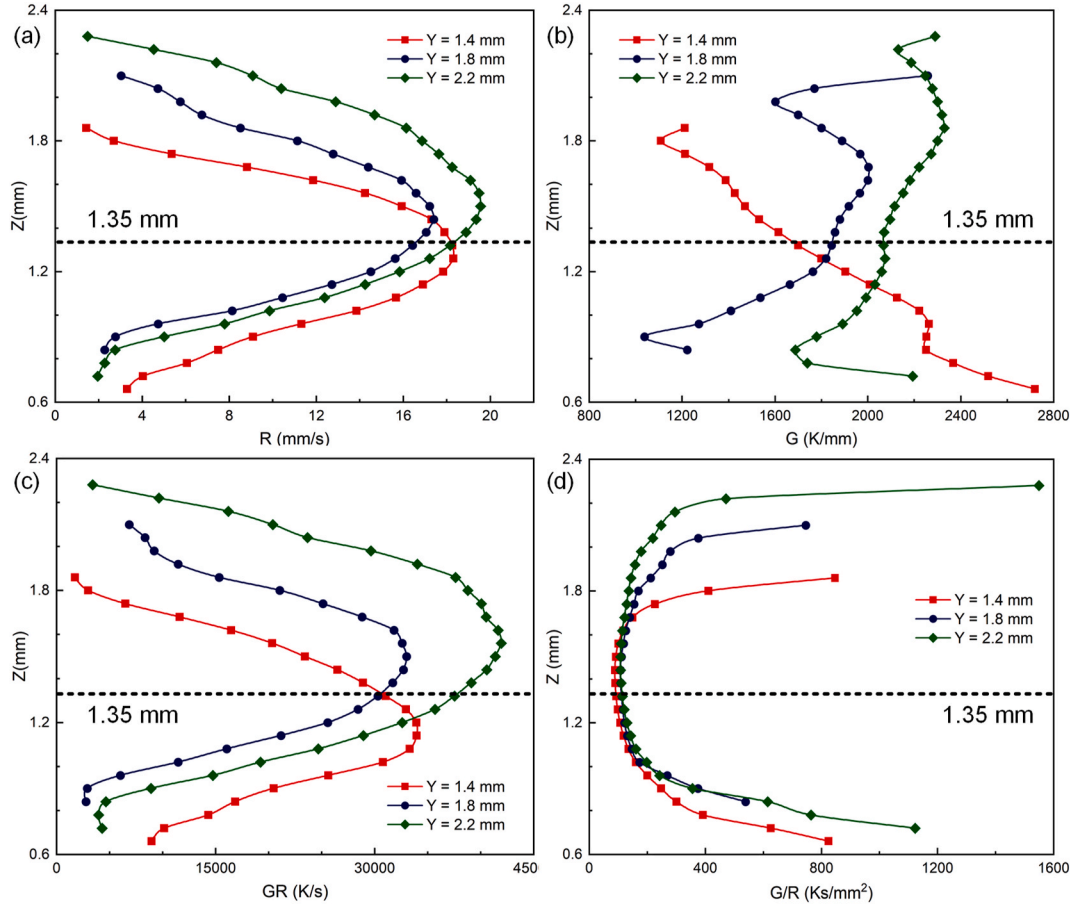


Fig. 14. Solidification parameters along isothermal (1783 K) at 80 ms: (a) solidification rate R ; (b) temperature gradient G ; (c) cooling rate GR ; (d) the degree of supercooling G/R .

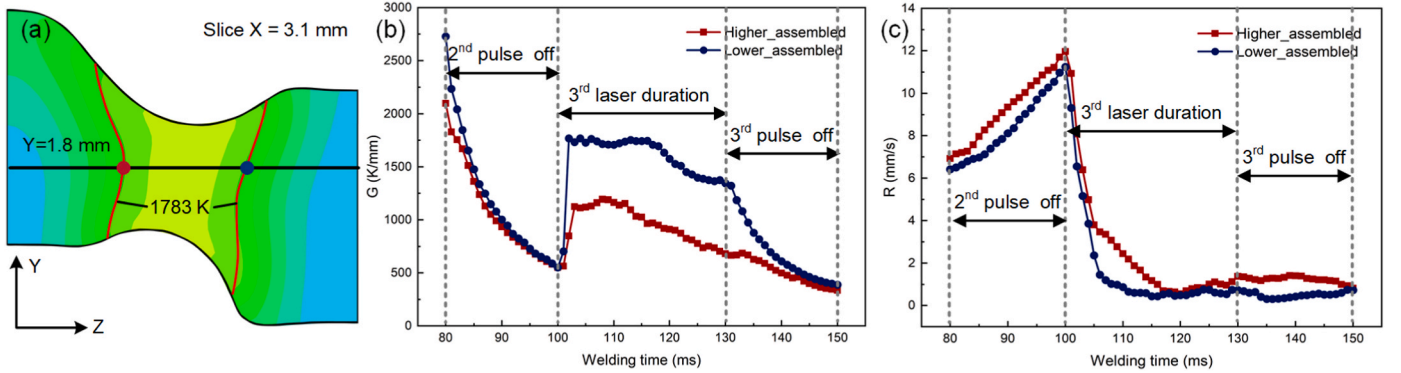


Fig. 15. Solidification parameters with welding time: (a) sampling locations; (b) G ; (c) R .

$X = 3.1$ mm (location of laser spot center at 80 ms), horizontal section $Y = 1.8$ mm (mid-thickness plane), and isotherm $T = 1783$ K (mean value of T_S and T_L). The variation curves along welding time are presented in Fig. 15. G steadily fell with the shrinking of the molten pool from 80 ms to 100 ms, corresponding to the second laser off stage, whereas R gradually increased. During the third pulse cycle, the section was reheated again. The temperature gradient G had a limited elevation over the laser radiation stage, reaching half of its value at 80 ms when solidification initially began, and then continuously declined throughout the pulse interval. At the meantime, the solidification rate R dropped rapidly to a quite low value and then fluctuated up and down. In particular, the asymmetric patterns of heat transfer and metal flow

caused by the assembly tolerance had more influence on the temperature gradient. G values on the lower-assembled side were bigger than those on the higher-assembled side, whereas R values on both sides did not differ much.

3.5. Assessment of butt joint with assembly tolerance from PLBW process

To evaluate the reliability of the PLBW process for connecting butt joints with assembly tolerances, the surface forming quality and mechanical properties of the weld bead were analyzed. Due to the misaligned edges, we inserted a pad with a thickness of 0.6 mm at each clamped end of the tensile specimen to ensure the tensile load was

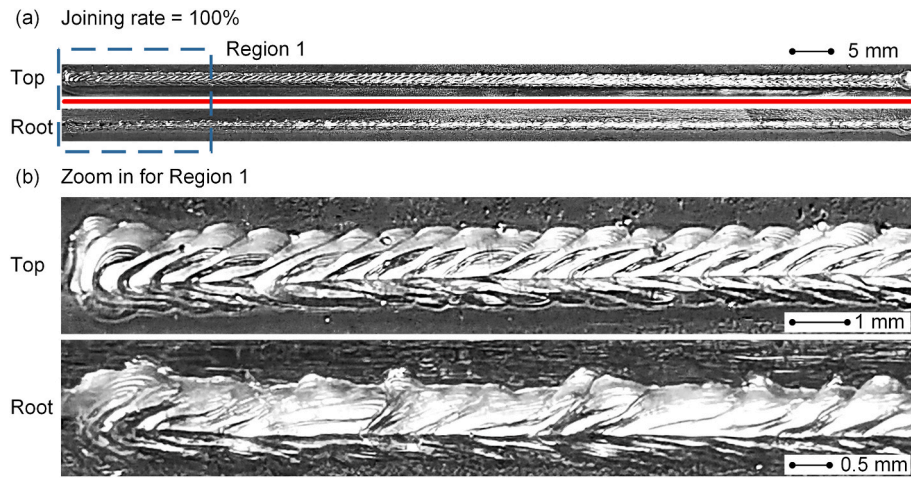


Fig. 16. Surface appearance of weld bead with manufacturing tolerance: (a) overall; (b) local zoom in.

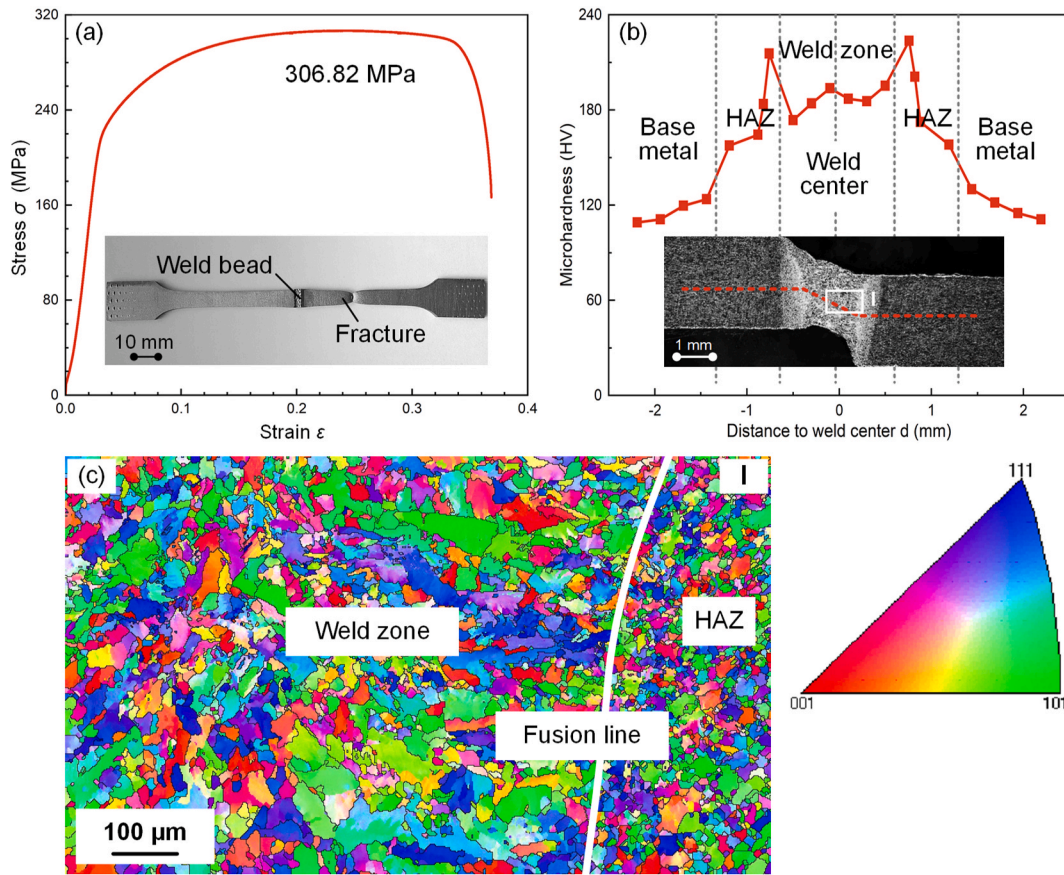


Fig. 17. Mechanical properties of weld bead with manufacturing tolerance: (a) stress-strain curve; (b) microhardness distribution; (c) EBSD image on weld cross-section.

applied in a straight line. Besides, The specimen dimensions and loading method were determined according to GB/T 2651-2008. For hardness measurement, the test points need to be carefully examined to distribute them along the mid-thickness line. Both the top and root surfaces of the weld exhibited continuous and uniform forming with a 100 % connection rate, as shown in Fig. 16. The specimen that experienced the tensile test fractured in the base material zone, indicating the strength of the weld seam was superior, as in Fig. 17(a). The micro-hardness in weld zone appeared to be greater than that in base metal, and the values were close to the experimental results from Meng et al. [46], depicted in

Fig. 17(b). Fig. 17(c) exhibited the EBSD image taken at the location of region I in Fig. 16(c). The weld region near the fusion line was arranged with columnar crystals in larger sizes, while the neighboring HAZ was distributed with fine equiaxed grains, resulting in greater hardness values. Moving from the fusion line to the weld center, equiaxed grains gradually emerged, but their sizes remained greater than in HAZ. In a nutshell, the PLBW technique is dependable for mild steel butt joints with assembly tolerances (an air gap of 0.3 mm as well as an edge misalignment of 0.6 mm).

4. Conclusions

Prior research has demonstrated the adaptability of laser-assisted welding techniques to assembly tolerances and the feasibility of CFD approaches for analyzing the kinetic behaviors of the molten pool. These studies, however, either require additional cost or do not focus on the dynamic weld pool and its solidification characteristics in the presence of both air gaps and edge misalignments. In this paper, butted steel plates with manufacturing tolerances were connected by the PLBW process, resulting in a well-formed and reliable joint. A multi-physics field coupling CFD model was developed to explore the joining mechanism. The characteristic solidification parameters in the molten pool with an air gap as well as an edge misalignment were estimated based on the melt flow and heat transfer. The predicted weld pool dimensions agreed well with the corresponding experimental outcomes. The following are the primary findings that can be derived from this work:

- The liquid metal at the rear part of weld pool first bulged and combined together under surface tension and recoil pressure. Then the unilateral bridge gradually converted into a bilateral state. The dominance of weld geometry by the higher-assembled side was broken with the formation of the liquid bridge. After the laser turned off, the two metal sheets were connected through the solidified bridge.
- The absorptivity of laser energy varied with the geometries of the molten pool and keyhole, and the absorbed laser energy thereafter spread predominantly by thermal convection.
- Because of edge misalignment, the liquid phase mass transfer rate across the mid-thickness plane displayed more prominent negative values consistent with gravity-driven flow. The top and root surfaces of the molten pool fluctuated at a characteristic frequency that was identical to the laser pulse frequency.
- The asymmetric patterns of heat transfer and fluid flow exerted a greater influence on temperature gradient than solidification rate. G did not present the typical distribution pattern reported in the literature while R was fairly consistent.

The results provide an economical and efficient tool to deal with manufacturing tolerances, while also filling a gap in the thermal-

fluid-metallurgical numerical analysis of butt joints in non-ideal assembly states. For those who are limited or no access to enhancing the fundamental laser welding system, this method is very convenient to implement by adjusting the laser pattern. In addition, the strategy of predicting weld formation based on heat-flow simulation is generic and practical. However, some limitations are worth noting. We must admit that our model has a memory limitation on the number of grids, necessitating extreme caution in determining the computational duration. In the future, we will make improvements in this area to break through the restriction.

Credit authors statement

Fengye Tang: Methodology, Investigation, Visualization, Writing – original draft. Kai Guo: Methodology, Investigation. Xiangbo Liu: Methodology, Writing – review & editing. Yanhong Wei: Conceptualization, Supervision. Jicheng Chen: Conceptualization, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Governing equations for laser welding [13,14]

- Molten pool free surface capture technology,

$$\frac{\partial(\rho_v H_v)}{\partial t} + \nabla \cdot (\rho_v H_v \mathbf{u}) = \dot{m}_{lv}. \quad (A1)$$

where, ρ is the density, H shows the VOF function, $\mathbf{u}$ means the velocity vector, $\dot{m}_{lv}$, expanded in Eq. (A2), represents the evaporation flux.

$$\text{for } T \geq T_v, \dot{m}_{lv} = \omega_{evap} H_l \rho_l \frac{T - T_v}{T_v} \quad (A2)$$

where, T is the temperature field, T_v is the boiling temperature, ω_{evap} indicates the evaporation rate, subscripts l and v mean the liquid and vapor phases, accordingly.

The VOF function H , built on LS function φ at the initialization step, ranges from zero to one, as given in Eq. (A3), where dx is the cell size.

$$H = \frac{1}{2} (1 + \tanh(\beta \varphi)), \beta = 0.85 dx. \quad (A3)$$

- Mass conservation equation,

$$\nabla \cdot \mathbf{u} = 0 \quad (A4)$$

- Momentum conservation equation,

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u u) = -\nabla p + \mu \nabla^2 u - K u + G + F_B + F_s + F_R, \quad (A5)$$

where, p is pressure field, μ is dynamic viscosity, K is drag coefficient in mushy zone, G is gravity force, F_B is buoyancy force, F_s is surface tension and F_R is recoil force in obedience to Eq. (A6).

$$F_R = 0.5 B_0 T^{-1/2} \exp(-m_v L_v / (k_b T_v)) N, \quad (A6)$$

where, B_0 is the evaporation constant, m_v is the molar mass, L_v is the latent heat of evaporation, k_b is the Boltzmann constant, and N is the normal vector of gas-liquid interface.

- Energy conservation equation,

$$\frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho h u) = \nabla \cdot (\lambda \nabla T) + S_{laser} - S_{evap}, \quad (A7)$$

where, λ is the thermal conductivity and h is the enthalpy of fluid given in Eq. (A8), with C_p the specific heat, f_{liq} the liquid phase volume fraction and L_m the latent heat of melting.

$$h = C_p T + f_{liq} L_m. \quad (A8)$$

The last two terms of Eq. (A7) exhibit the source term from laser input and evaporation loss.

$$S_{laser} = Q_{Fa} + Q_p, \quad (A9)$$

$$S_{evap} = \dot{m}_v L_v, \quad (A10)$$

where, Q_{Fa} is the energy from Fresnel absorption, Q_p is the energy from superficial plume heating.

References

- [1] H. Wan, H. Cao, S. Luan, C. Gui, Investigation of thermal damage in continuous wave laser-induced nanowelding, *Opt Laser. Technol.* 161 (2023), 109143, <https://doi.org/10.1016/j.optlastec.2023.109143>.
- [2] H. Fan, A. Witvrouw, F. Wolf-monheim, R. Souschek, S. Yang, Effects of substrate surface treatments on hybrid manufacturing of AlSi7Mg using die casting and selective laser melting, *J. Mater. Sci. Technol.* 156 (2023) 142–156, <https://doi.org/10.1016/j.jmst.2023.02.009>.
- [3] J. Li, W. Zhang, H. Zheng, J. Gao, C. Jiang, Reducing plasma shielding effect for improved nanosecond laser drilling of copper with applied direct current, *Opt Laser. Technol.* 163 (2023), 109372, <https://doi.org/10.1016/j.optlastec.2023.109372>.
- [4] Z.M. Yang, H.G. Yan, J.H. Chen, B. Su, G.H. Zhang, Q. Zhao, H.G. Yan, J.H. Chen, B. Su, G.H. Zhang, Q. Zhao, Microstructural characterisation and liquation behaviour of laser welded joint of fine grained AZ91 magnesium alloy thin sheets, *Sci. Technol. Weld. Join.* ISSN. 20 (2015) 27–34, <https://doi.org/10.1179/1362171814Y.00000000252>.
- [5] A. Yadav, J. Joshi, A. Chakraborty, H. Natsu, Laser welded lip seal for UHV class fusion vessels – a methodical study, *Fusion Eng. Des.* 180 (2022), 113163, <https://doi.org/10.1016/j.fusengdes.2022.113163>.
- [6] M.R. Ömer Üstündağ, Nasim Bakir, Sergej Gook, Andrey Gumenyuk, Hybrid laser-arc welding of laser- and plasma-cut 20-mm-thick structural steels, *Weld. World* 66 (2022) 507–514, <https://doi.org/10.1007/s40194-022-01255-y>.
- [7] M.S.S. Egerland, H. Staufer, M. Ruehrnoess, Investigation of advanced laser-MAG tandem hybrid welding for joining gap-flawed thin sheet metal parts, *Weld. World* 62 (2018) 95–104, <https://doi.org/10.1007/s40194-017-0531-0>.
- [8] Z. Sun, M. Kuo, Bridging the joint gap with wire feed laser welding, *J. Mater. Process. Technol.* 87 (1999) 213–222, [https://doi.org/10.1016/S0924-0136\(98\)00346-x](https://doi.org/10.1016/S0924-0136(98)00346-x).
- [9] J. Pu, Y. Zhao, Y. Jiang, J. Zou, S. Song, Influence of wire feeding speed on the melting behavior and formation of narrow-gap joint by laser welding with filler wire, *J. Laser Appl.* 32 (2021), 032007, <https://doi.org/10.2351/7.0000043>.
- [10] C.W. Yang Tao, Junfeng Liu, Zhuang Yuan, Kai Sun, Studies on the formation mechanism of incomplete fusion defects in ultra- narrow gap laser wire filling welding, *Opt Laser. Technol.* 129 (2020), 106275, <https://doi.org/10.1016/j.optlastec.2020.106275>.
- [11] V. Schultz, T. Seefeld, F. Vollertsen, Gap bridging ability in laser beam welding of thin aluminum sheets, *Phys. Procedia* 56 (2014) 545–553, <https://doi.org/10.1016/j.phpro.2014.08.037>.
- [12] H. J. K.D. Markus Köhler, Tamás tóth, andreas kreybohm, effects of reduced ambient pressure and beam oscillation on gap bridging ability during solid-state laser beam welding, *J. Manuf. Mater. Process. Artic.* 4 (2020) 40, <https://doi.org/10.3390/jmmp4020040>.
- [13] J. Chen, Z. Ouyang, X. Du, Y. Wei, Weld pool dynamics and joining mechanism in pulse wave laser beam welding of Ti-6Al-4V titanium alloy sheets assembled in butt joint with an air gap, *Opt Laser. Technol.* 146 (2022), 107558, <https://doi.org/10.1016/j.optlastec.2021.107558>.
- [14] F. Tang, Y. Wei, L. Qian, Y. Tang, J. Chen, Asymmetry of keyhole and weld pool geometry in PLBW of tailor-welded steel sheets with edge misalignment : numerical modeling and experimental validation, *Opt Laser. Technol.* 161 (2023), 109205, <https://doi.org/10.1016/j.optlastec.2023.109205>.
- [15] Y.F. Tzeng, Process characterization of pulsed Nd:YAG laser seam welding, *Int. J. Adv. Manuf. Technol.* 16 (2000) 10–18, <https://doi.org/10.1007/PL00013126>.
- [16] M.J. Torkamany, M.J. Hamed, F. Malek, J. Sabbaghzadeh, The effect of process parameters on keyhole welding with a 400 W Nd:YAG pulsed laser, *J. Phys. D Appl. Phys.* 39 (2006) 4563–4567, <https://doi.org/10.1088/0022-3727/39/21/009>.
- [17] Z. Mohid, N.H. Rafai, M. Rasidi Ibrahim, E.A. Rahim, Melted zone shapes transformation in titanium alloy welded using pulse wave laser, *Mater. Sci. Forum* 882 (2017) 8–12, <https://dx.doi.org/10.4028/www.scientific.net/MSF.882.8>.
- [18] E. Assuncao, S. Williams, Comparison of continuous wave and pulsed wave laser welding effects, *Opt Laser. Eng.* 51 (2013) 674–680, <https://doi.org/10.1016/j.optlaseng.2013.01.007>.
- [19] M.J. Torkamany, S. Tahamtan, J. Sabbaghzadeh, Dissimilar welding of carbon steel to 5754 aluminum alloy by Nd:YAG pulsed laser, *Mater. Des.* 31 (2010) 458–465, <https://doi.org/10.1016/j.matdes.2009.05.046>.
- [20] P. von Witzendorff, J. Hermsdorf, S. Kaierle, O. Suttman, L. Overmeyer, Double pulse laser welding of 6082 aluminium alloys, *Sci. Technol. Weld. Join.* 20 (2015) 42–47, <https://doi.org/10.1179/1362171814Y.00000000255>.
- [21] M. Chludzinski, R.E. dos Santos, C. Churique, S.R. Fernández-Vidal, M. Ortega-Iguña, J.M. Sánchez-Amaya, Pulsed laser butt welding of AISI 1005 steel thin plates, *Opt Laser. Technol.* 134 (2021), 106583, <https://doi.org/10.1016/j.optlastec.2020.106583>.
- [22] X.L. Gao, L.J. Zhang, J. Liu, J.X. Zhang, Porosity and microstructure in pulsed Nd:YAG laser welded Ti6Al4V sheet, *J. Mater. Process. Technol.* 214 (2014) 1316–1325, <https://doi.org/10.1016/j.jmatprotec.2014.01.015>.
- [23] J.H. Cho, S.J. Na, Implementation of real-time multiple reflection and Fresnel absorption of laser beam in keyhole, *J. Phys. D Appl. Phys.* 39 (2006) 5372–5378, <https://doi.org/10.1088/0022-3727/39/24/039>.
- [24] D. Wu, X. Hua, Y. Ye, L. Huang, Experimental and numerical study of spatter formation and composition change in fiber laser welding of aluminum alloy, *J. Phys. D Appl. Phys.* 51 (2018), 185604, <https://doi.org/10.1088/1361-6463/aab758>.
- [25] W. Cho, V. Schultz, P. Woizeschke, Numerical study of the effect of the oscillation frequency in buttonhole welding, *J. Mater. Process. Technol.* 261 (2018) 202–212, <https://doi.org/10.1016/j.jmatprotec.2018.05.024>.
- [26] W. Cho, P. Woizeschke, Analysis of molten pool dynamics in laser welding with beam oscillation and filler wire feeding, *Int. J. Heat Mass Tran.* 164 (2021), 120623, <https://doi.org/10.1016/j.ijheatmasstransfer.2020.120623>.

- [27] W. Ke, Z. Zeng, J.P. Oliveira, B. Peng, J. Shen, C. Tan, X. Song, W. Yan, Heat transfer and melt flow of keyhole, transition and conduction modes in laser beam oscillating welding, *Int. J. Heat Mass Tran.* 123 (2023) 123821, <https://doi.org/10.1016/j.ijheatmasstransfer.2022.123821>.
- [28] L. Li, G. Peng, J. Wang, J. Gong, S. Meng, Numerical and experimental study on keyhole and melt flow dynamics during laser welding of aluminium alloys under subatmospheric pressures, *Int. J. Heat Mass Tran.* 133 (2019) 812–826, <https://doi.org/10.1016/j.ijheatmasstransfer.2018.12.165>.
- [29] S. Huang, L. Xu, M. Lou, H. Chen, K. Zhang, Y. Li, Keyhole-induced pore formation mechanism in laser-MIG hybrid welding of aluminum alloy based on experiment and multiphase numerical model, *J. Mater. Process. Technol.* 314 (2023) 117903, <https://doi.org/10.1016/j.jmatprotec.2023.117903>.
- [30] H.L. Wei, J. Mazumder, T. Debroy, Evolution of solidification texture during additive manufacturing, *Nat. Publ. Gr.* (2015) 1–7, <https://doi.org/10.1038/srep16446>.
- [31] W. Ou, T. Mukherjee, G.L. Knapp, Y. Wei, T. Debroy, Fusion zone geometries, cooling rates and solidification parameters during wire arc additive manufacturing, *Int. J. Heat Mass Tran.* 127 (2018) 1084–1094, <https://doi.org/10.1016/j.ijheatmasstransfer.2018.08.111>.
- [32] S. Geng, P. Jiang, X. Shao, L. Guo, X. Gao, Heat transfer and fluid flow and their effects on the solidification microstructure in full-penetration laser welding of aluminum sheet, *J. Mater. Sci. Technol.* 46 (2020) 50–63, <https://doi.org/10.1016/j.jmst.2019.10.027>.
- [33] Z. Li, G. Yu, X. He, S. Li, C. Tian, B. Dong, Analysis of surface tension driven flow and solidification behavior in laser linear welding of stainless steel, *Opt. Laser. Technol.* 123 (2020), 105914, <https://doi.org/10.1016/j.optlastec.2019.105914>.
- [34] T.T. Roehling, R. Shi, S.A. Khairallah, J.D. Roehling, G.M. Guss, J.T. Mckeown, M. J. Matthews, Controlling grain nucleation and morphology by laser beam shaping in metal additive manufacturing, *Mater. Des.* 195 (2020), 109071, <https://doi.org/10.1016/j.matdes.2020.109071>.
- [35] J. Zhao, P. Jiang, S. Geng, L. Guo, Y. Wang, Experimental and numerical study on the effect of increasing frequency on the morphology and microstructure of aluminum alloy in laser wobbling welding, *J. Mater. Res. Technol.* 21 (2022) 267–282, <https://doi.org/10.1016/j.jmrt.2022.09.008>.
- [36] H. Hao, J. Gao, H. Huang, Numerical simulation for dynamic behavior of molten pool in tungsten inert gas welding with reserved gap, *J. Manuf. Process.* 58 (2020) 11–18, <https://doi.org/10.1016/j.jmapro.2020.07.063>.
- [37] Y. Feng, X. Gao, Y. Zhang, C. Peng, X. Gui, Y. Sun, X. Xiao, Simulation and experiment for dynamics of laser welding keyhole and molten pool at different penetration status, *Int. J. Adv. Manuf. Technol.* (2021) 2301–2312, <https://doi.org/10.1007/s00170-020-06489-y>.
- [38] J. Chen, X. Chen, X. Liu, Y. Wei, Numerical investigation on keyhole collapsing and rebuilding behavior during pulsed laser beam welding of Ti6Al4V titanium alloy under various pulse frequencies, *Appl. Phys. Mater. Sci. Process* 128 (2022) 1–15, <https://doi.org/10.1007/s00339-021-05006-6>.
- [39] W. Ge, J.Y.H. Fuh, S.J. Na, Numerical modelling of keyhole formation in selective laser melting of Ti6Al4V, *J. Manuf. Process.* 62 (2021) 646–654, <https://doi.org/10.1016/j.jmapro.2021.01.005>.
- [40] T.R. Allen, W. Huang, J.R. Tanner, W. Tan, J.M. Fraser, B.J. Simonds, Energy-coupling mechanisms revealed through simultaneous keyhole depth and absorptance measurements during laser-metal processing, *Phys. Rev. Appl.* 13 (2020) 1, <https://doi.org/10.1103/physrevapplied.13.064070>.
- [41] R. Rai, S.M. Kelly, R.P. Martukanitz, T. DebRoy, A convective heat-transfer model for partial and full penetration keyhole mode laser welding of a structural steel, *Metall. Mater. Trans. A Phys. Metall. Mater. Sci.* 39 (2008) 98–112, <https://doi.org/10.1007/s11661-007-9400-6>.
- [42] J. Chen, X. Du, B. Kong, Y. Wei, Effects of groove clearance size on gap bridging capacity in PWLBW of Ti6Al4V alloy sheet assembled in butt joint configuration: numerical simulation and experimental assessment, *Opt. Laser. Technol.* 156 (2022), 108527, <https://doi.org/10.1016/j.optlastec.2022.108527>.
- [43] H. Li, Y. Huang, S. Jiang, Y. Lu, X. Gao, X. Lu, Z. Ning, J. Sun, Columnar to equiaxed transition in additively manufactured CoCrFeMnNi high entropy alloy, *Mater. Des.* 197 (2021), 109262, <https://doi.org/10.1016/j.matdes.2020.109262>.
- [44] G.L. Knapp, N. Raghavan, A. Plotkowski, T. DebRoy, Experiments and simulations on solidification microstructure for Inconel 718 in powder bed fusion electron beam additive manufacturing, *Addit. Manuf.* 25 (2019) 511–521, <https://doi.org/10.1016/j.addma.2018.12.001>.
- [45] Y. Hou, Z. Ren, Z. Zhang, X. Ren, Columnar to equiaxed transition during directionally solidifying GCr18Mo steel affected by thermoelectric magnetic force under an axial static magnetic field, *ISIJ Int.* 59 (2019) 60–68, <https://doi.org/10.2355/isijinternational.ISIJINT-2018-449>.
- [46] X. Meng, G. Qin, Y. Zhang, B. Fu, Z. Zou, Journal of Materials Processing Technology High speed TIG – MAG hybrid arc welding of mild steel plate, *J. Mater. Process. Technol.* 214 (2014) 2417–2424, <https://doi.org/10.1016/j.jmatprotec.2014.05.020>.